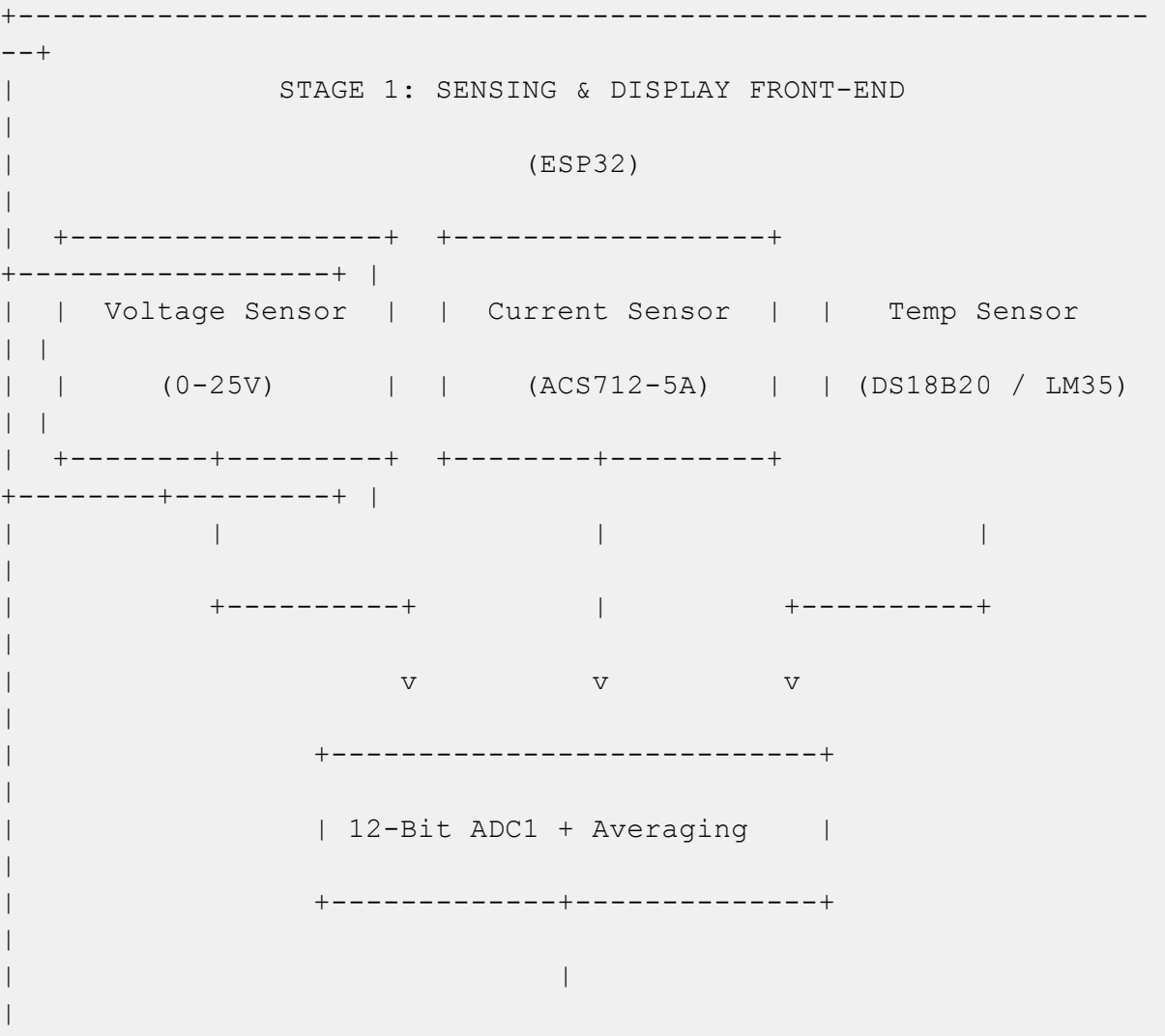
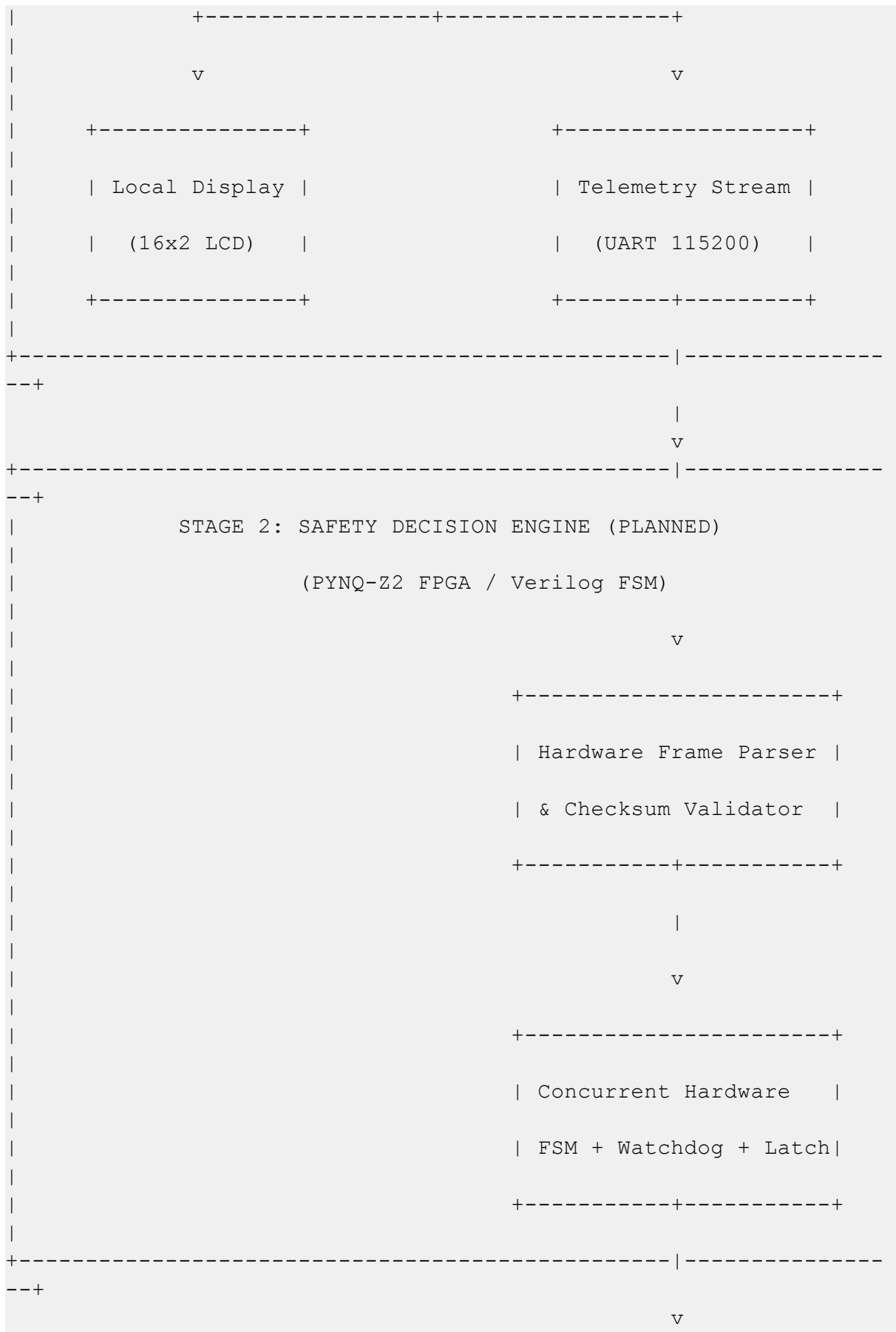


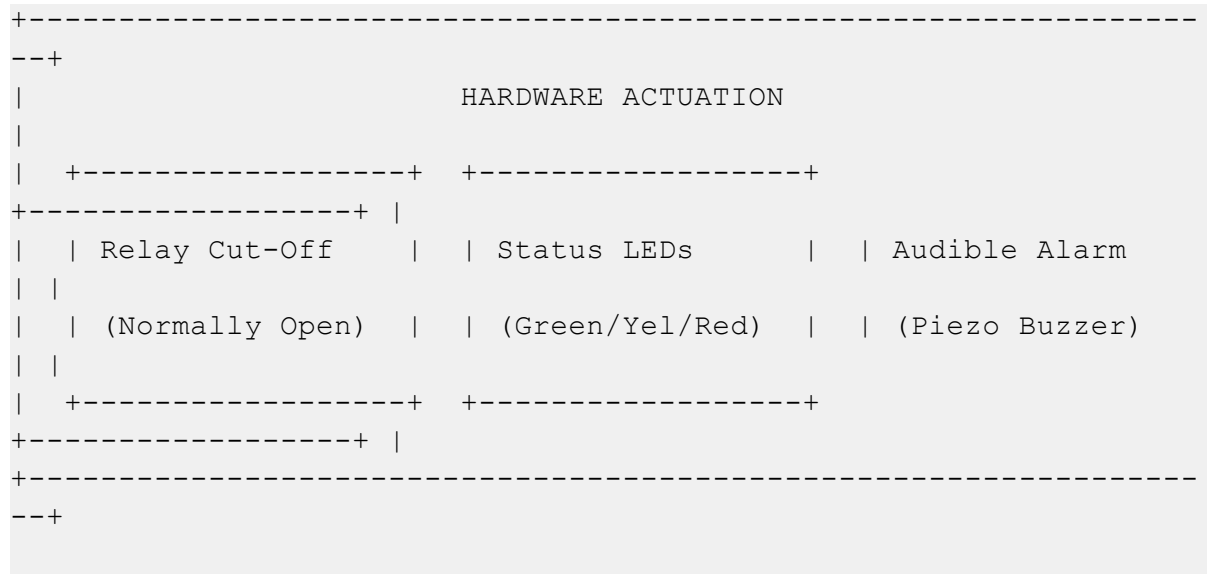
Engineering Specification: ESP32 Hardware Safety Layer & Interface Protocol

1. System Architecture & Fail-Safe Design Philosophy

Lithium-ion battery chemistries deployed in mobile devices and portable electronics exhibit high energy density but suffer from severe vulnerability to thermal runaway, cell swelling, mechanical venting, and fire when subjected to over-voltage, over-current, or elevated ambient operating temperatures. Conventional battery management routines operating within software-driven operating systems present an unacceptable single point of failure; an OS kernel panic, scheduler delay, or unhandled software exception can freeze monitoring routines and leave charging paths energized during an anomaly. To guarantee non-negotiable functional safety, an independent, hardware-level safety layer situated directly between the primary power source and the cell load is mandatory. Operating independently of primary system software, this safety layer performs real-time parametric threshold evaluation to enforce immediate electrical isolation of the charging circuit before cell degradation or catastrophic thermal degradation occurs. The system relies on a dual-stage architecture that strictly decouples analog parameter acquisition from safety-critical decision evaluation:







1. **Sensing and Display Front-End (ESP32):** Performs high-resolution 12-bit analog sampling across dedicated ADC1 channels, executes digital filtering via sample averaging, drives a local 16x2 I2C character LCD, and formats normalized parameters into digital frames. In the **Verified Current Prototype Build**, the ESP32 evaluates threshold logic locally in firmware and directly drives output actuators (Relay on GPIO26, Buzzer on GPIO25, and LEDs on GPIO2/4/15). In the **Planned Target Architecture**, the ESP32 serves purely as a front-end telemetry transmitter, streaming a 5-byte packet over a 115200-baud UART interface to the secondary hardware engine while maintaining local display fallback.
2. **Safety Decision Engine (PYNQ-Z2 FPGA / Target Logic):** Serves as the upstream target decision layer in the planned deployment. Operating in Verilog fabric on a Zynq-7000 FPGA, this stage receives the UART telemetry stream, validates frame integrity via an XOR checksum, and evaluates parameters against hardware comparators concurrently every clock cycle. Offloading threshold evaluation to hardware logic creates a deterministic, parallel safety path completely free of operating system overhead, interrupt latency, or thread starvation risks. To achieve robust fault tolerance, the hardware isolation path relies strictly on a fail-safe execution principle. The power-path switching element utilizes the Normally Open (NO) contacts of an electromechanical relay module. To maintain power delivery to the charging load, the active safety engine must continuously supply a logical high control signal (HIGH / 3.3 V) to energize the relay coil. Any system anomaly—including complete loss of supply voltage, microcontroller code execution stall, UART frame corruption, wire disconnection, or hardware watchdog expiration—immediately de-energizes the relay coil. Mechanical spring return forces the contacts open within milliseconds, dropping the load and isolating the power source. To prevent ambiguity between current capabilities and planned deployment features, the system specifications are partitioned into verified prototype parameters and target production specifications:

Architecture	Feature	Verified Current Prototype Build	Planned Target Architecture
Parameter Input Source	3x $10\text{ k}\Omega$ Potentiometers (POT1, POT2, POT3 simulating 0–100% full-scale ranges)	✓	✓
	Physical Transducers (0–25V Voltage Module, ACS712-5A Current Sensor, DS18B20 / LM35 Sensor)	✓	✓
Thermal Sensing Engine	DS18B20 Digital 1-Wire Sensor on GPIO19 ($4.7\text{ k}\Omega$ pull-up to 3.3V)	✓	✓

Cell-mounted surface thermal sensor array (LM35 analog or DS18B20 digital) || **Logic & Actuation Control** | Local ESP32 software threshold evaluation driving GPIO26 (Relay) directly | Streamed 115200-baud UART telemetry driving a Verilog FSM on a PYNQ-Z2 FPGA || **State Machine Execution** | Sequential conditional logic executing in ESP32 firmware loop | Parallel hardware FSM in FPGA fabric with watchdog, hysteresis, and latched fault recovery || **Telemetry Protocol** | Local 16x2 I2C LCD drive (Address 0x27) and direct GPIO status LED drive | Dual-stage: Local LCD backup + 5-byte UART telemetry packet (0xA5 sync byte + XOR checksum) || **Fault Recovery Protocol** | Real-time state clearing based on immediate analog potentiometer wiper position | Latched fault state requiring a physical reset button pulse once parameters fall below hysteresis bands || **Watchdog Cut-Off** | N/A (Local synchronous execution loop) | Hardware 1000 ms timer; missing/corrupted UART frames force immediate load drop || **Hysteresis Band** | N/A (Direct software threshold comparison) | 5% (5-point) deadband below entry boundaries to eliminate relay chatter at limit transitions || **Power-On Boot Logic** | Relay defaults to de-energized; energized upon firmware state machine initialization | Relay remains de-energized (LOW) until first valid frame confirms an explicit SAFE condition |

With the high-level dual-stage safety architecture and fail-safe operating philosophy established, realizing this system requires strict adherence to precise hardware pin allocations.

2. Master ESP32 Pin Allocation & Interface Mapping

Implementing an embedded hardware safety layer on the ESP32 micro-controller requires careful consideration of silicon-level pin constraints. Hardware pin mapping must strictly avoid bus contention, account for internal boot-strapping logic, maintain separation between low-power analog sensing and high-current switching paths, and restrict analog sampling exclusively to ADC1 peripherals to eliminate conflicts with Wi-Fi/system calls. The table below documents the master pin allocations verified in the current physical build:

GPIO Pin	Hardware Function / Subsystem	Physical Wiring Connection / Schematic Details	Verified Logic Level / Operating Mode
GPIO32	Voltage Sensor Input (S)	Direct connection to Signal (S) pin of 0–25V module Analog Input (ADC1 Channel, 0 V)	---
GPIO35	ACS712 Current Sensor	Connected to central measurement node of $10\text{ k}\Omega / 20\text{ k}\Omega$ divider Analog Input (ADC1 Channel, Input-Only Pin)	---
GPIO33	POT1 Voltage Simulation	Connected to wiper terminal of Potentiometer 1 (Voltage simulator) Analog Input (ADC1 Channel, 0 V swing)	---
GPIO36 (VP)	POT2 Current Simulation	Connected to wiper terminal of Potentiometer 2 (Current simulator) Input-Only Analog (ADC1 Channel, No Internal Pull-Up/Down)	---
GPIO39 (VN)	POT3 Temp Simulation	Connected to wiper terminal of Potentiometer 3 (Temp simulator / LM35) Input-Only Analog (ADC1 Channel, No Internal Pull-Up/Down)	---
GPIO19	DS18B20 Digital Temp	Connected to DS18B20 DATA line; $4.7\text{ k}\Omega$ pull-up tied to 3.3 V rail Digital Input / Output (1-Wire Bidirectional Protocol)	---
GPIO27	START Push Button	Switch contact connected to GPIO27; opposite terminal to Common GND INPUT_PULLUP (Digital Input, Active-LOW when pressed)	---
GPIO14	RESET Push Button	Switch contact connected to GPIO14; opposite terminal to Common GND INPUT_PULLUP (Digital Input, Active-LOW when pressed)	---
GPIO18	FAULT TEST Button	Switch contact connected to GPIO18; opposite terminal to Common GND Digital Input (Active-LOW when pressed)	---
GPIO23	MENU Push Button	Switch contact connected to GPIO23; opposite terminal to Common GND	---

INPUT_PULLUP (Digital Input, Active-LOW when pressed) || **GPIO26** | Relay Module Control | Direct connection to Relay Module IN signal input terminal | Digital Output (HIGH = Relay Energized / Closed) || **GPIO25** | Piezo Alarm Control | Direct connection to Piezo Buzzer positive (+) terminal | Digital Output (HIGH = Buzzer Active / Alarm On) || **GPIO2** | Green Status LED | Connected via series $220\text{ }\Omega$ resistor to LED Anode; Cathode to GND | Boot-Strapping Pin / Digital Output (HIGH = SAFE) || **GPIO15** | Red Status LED | Connected via series $220\text{ }\Omega$ resistor to LED Anode; Cathode to GND | Boot-Strapping Pin / Digital Output (HIGH = FAULT) || **GPIO4** | Yellow Status LED | Connected via series $220\text{ }\Omega$ resistor to LED Anode; Cathode to GND | Standard Digital Output (HIGH = WARNING) || **GPIO21** | I2C Serial Data (SDA) | Connected directly to 16x2 LCD I2C Backpack SDA terminal | Open-Drain Digital I/O (I2C Address 0x27) || **GPIO22** | I2C Serial Clock (SCL) | Connected directly to 16x2 LCD I2C Backpack SCL terminal | Open-Drain Digital Output (I2C Bus Clock) |

Strapping Pin & Floating Input Analysis

1. **Silicon Boot Strapping Mechanics (GPIO2 & GPIO15):** GPIO2 and GPIO15 function as critical boot-configuration pins during power-on reset (POR). The internal state of GPIO2 determines whether the ESP32 enters standard flash boot execution or UART download mode, while GPIO15 controls silent boot logging. Holding these pins in an incorrect logic state during power application will lock up the processor core and halt system boot. To prevent low-impedance external components from pulling these nodes down at start-up, dedicated $220\text{ }\Omega$ current-limiting resistors are installed in series between the GPIO pins and the status LED anodes. These resistors provide isolation that allows internal pull resistors to register correct boot voltages while maintaining nominal current drive ($\sim 7.7\text{ mA}$) for status illumination during active operation.
2. **ADC Peripheral Allocation (ADC1 vs. ADC2):** All analog monitoring signals (GPIO32, GPIO33, GPIO35, GPIO36, GPIO39) are assigned exclusively to **ADC1** . Silicon routing restrictions on the ESP32 dictate that **ADC2** pins cannot perform analog-to-digital conversions when internal Wi-Fi, Bluetooth, or specific system driver tasks are active. Assigning parameter acquisition strictly to ADC1 guarantees unhindered analog conversion capability under all firmware operating conditions.
3. **Input-Only Pin Constraints (GPIO36 / VP & GPIO39 / VN):** GPIO36 and GPIO39 are strictly input-only pins on the ESP32 architecture. They lack internal pull-up or pull-down transistor arrays within the silicon matrix. Consequently, these pins cannot be configured via software using INPUT_PULLUP mode. Signal conditioning networks tied to GPIO36 and GPIO39 must incorporate external hardware termination to prevent floating node conditions when signal drivers are disconnected.
4. **Floating Node Elimination:** Tactile switches tied to GPIO27, GPIO14, GPIO18, and GPIO23 are set to INPUT_PULLUP mode in firmware (activating internal $\approx 45\text{ k}\Omega$ pull-up resistors to 3.3 V). When the switch is open, the input node is held high. When depressed, the contact creates a low-impedance path directly to Common GND, driving the pin LOW. This eliminates the need for external physical pull-down resistors while preventing floating inputs caused by high-impedance pickup. Having mapped the master pin allocations and electrical constraints, attention must be directed to the dedicated signal conditioning networks required for accurate analog sensor acquisition.

3. Sensor Interfaces & Signal Conditioning Networks

The integrated 12-bit Analog-to-Digital Converter (ADC) of the ESP32 measures voltages strictly within a 0 V to 3.3 V full-scale input window. Exposing an ADC pin to potential levels exceeding $V_{DD} + 0.3\text{ V}$ (3.6 V) forward-biases the internal ESD protection diodes, injecting substrate currents that can degrade adjacent ADC channels or cause permanent chip failure. Signal conditioning networks attenuate, level-shift, and filter transducer outputs to match the ESP32 input specifications.

VOLTAGE SENSING NETWORK (0-25V Module):

```
Source (+) ----- [Screw Terminal +] ---\
                                           |-- Internal 5:1
Attenuation --> S Pin ---> ESP32 GPIO32
Source (-) ----- [Screw Terminal -] ---/   (Header '+'
Unconnected)      Header '-' -> Common GND
```

CURRENT SENSING NETWORK (ACS712-5A):

```
ACS712 OUT (5V Supply, 2.5V Zero-Current Offset)
|
[10 kΩ Series Resistor]
|
+---> Measurement Node ---> ESP32 GPIO35 (ADC1 Input)
|
[20 kΩ Shunt Resistor]
|
Common GND
```

THERMAL MONITORING NETWORK (DS18B20 Current Build):

```
3.3V Power Rail -----+
                        |
                        [4.7 kΩ Pull-Up]
                        |
ESP32 GPIO19 <--- DATA Line -----+---> DS18B20 DATA Pin
```

Voltage Sensing Subsystem (0-25V Module)

The 0-25V voltage sensing module utilizes an onboard $5:1$ resistor divider network ($30\text{ k}\Omega$ series resistor paired with a $7.5\text{ k}\Omega$ shunt resistor).

- **Network Attenuation Mathematics:** The voltage attenuation ratio α_v is calculated as: $\alpha_v = \frac{7.5\text{ k}\Omega}{30\text{ k}\Omega + 7.5\text{ k}\Omega} = \frac{7.5}{37.5} = 0.2 = \frac{1}{5}$. The signal voltage delivered to the ADC pin (V_{ADC}) as a function of terminal input voltage (V_{in}) is expressed as: $V_{\text{ADC}} = \frac{V_{\text{in}}}{5}$. To reconstruct the absolute source voltage (V_{in}) in software or Verilog logic, the inversion equation is applied: $V_{\text{in}} = 5 \times V_{\text{ADC}}$. For a maximum charging bus potential of 16.0 V , the voltage present at GPIO32 is 3.20 V , positioning the measurement safely within the 3.3 V scale limits.
- **Wiring Topology & Header Mandate:** High-current power conductors from the charging adapter attach to the terminal block (Screw + to Source Positive, Screw - to

Source Return). Signal pin S connects directly to GPIO32, while Header - attaches to Common GND. Header pin + must remain strictly **unconnected**. Placing additional external attenuation resistors on top of the module's onboard network distorts the transfer function, introducing uncalibrated measurement errors.

Current Sensing Subsystem (ACS712-5A)

The ACS712-5A Hall-effect current sensor requires a nominal 5.0 V supply rail. Its internal architecture generates an output voltage centered at $V_{CC} / 2$ (2.50 V at zero current) with a proportional transfer sensitivity of 185 mV/A . At its positive full-scale limit of $+5.0\text{ A}$, the raw output reaches: $V_{\text{ACS_MAX}} = 2.50\text{ V} + (5.0\text{ A}) \times 0.185\text{ V/A} = 3.425\text{ V}$. Because 3.425 V exceeds the maximum operating limit of the ESP32 ADC, an external resistor divider network ($10\text{ k}\Omega$ series resistor and $20\text{ k}\Omega$ shunt resistor) is installed between the sensor output pin and Common GND.

- Signal Attenuation & Transfer Function Mathematics:** The voltage division ratio α_c scales the raw sensor output by: $\alpha_c = \frac{20\text{ k}\Omega}{10\text{ k}\Omega + 20\text{ k}\Omega} = \frac{2}{3}$. The attenuated voltage present at GPIO35 (V_{ADC}) is related to raw sensor output (V_{out}) by: $V_{\text{ADC}} = \frac{2}{3} \times V_{\text{out}}$ implies $V_{\text{out}} = 1.5 \times V_{\text{ADC}}$. Substituting the ACS712 physical current response ($V_{\text{out}} = 2.50\text{ V} + 0.185 \cdot I$) yields the complete system transfer equation: $1.5 \times V_{\text{ADC}} = 2.50\text{ V} + (0.185\text{ V/A}) \cdot I$. Solving directly for physical load current I in firmware or target FPGA logic: $I = \frac{1.5 \cdot V_{\text{ADC}} - 2.50\text{ V}}{0.185\text{ V/A}}$. This conditioning scales the full-scale output range ($0\text{ V} \text{--} 3.425\text{ V}$) down to $0\text{ V} \text{--} 2.283\text{ V}$ at GPIO35, preserving headroom below the 3.3 V ceiling.
- Low-Voltage Current Path Routing:** High-current charging lines must route directly through the low-resistance copper conduction path: Source + connects to sensor terminal IP+, IP- connects to Load +, and Load - returns directly to Source -. Full quantitative accuracy requires two-point zero-current offset calibration in software prior to operational use.

Thermal Monitoring Subsystem (DS18B20 vs. Legacy LM35)

- Verified DS18B20 Digital Integration:** The current working prototype utilizes a DS18B20 digital temperature sensor operating over a 1-Wire bus. The Red wire connects to the 3.3 V power rail, the Black wire attaches to Common GND, and the Yellow/White DATA line connects to GPIO19. A dedicated $4.7\text{ k}\Omega$ pull-up resistor must be tied between GPIO19 (DATA) and the 3.3 V rail to ensure bus idle stability. Under no circumstances may the DATA line be pulled up to 5.0 V .
- Legacy/Planned LM35 Analog Integration Note:** The original target architecture specification called for an LM35 precision analog temperature sensor delivering a linear output response of $10\text{ mV}/^{\circ}\text{C}$ (0.60 V at 60.0°C). When configured, the LM35 output connects directly to GPIO39 (ADC1) paired with a parallel 100 nF filtering capacitor. During

bench testing, severe analog offset drift caused by missing ground isolation resulted in thermal stress that damaged the initial LM35 component. The digital DS18B20 was integrated into the verified build to eliminate analog noise pickup on the thermal monitoring line, though the LM35 remains a valid configuration option for the target FPGA implementation provided proper star grounding is maintained.

Fault Simulation Network (3-Potentiometer Array)

For safe bench-top verification without exposing physical components to over-voltage or thermal stress, three $10\text{ k}\Omega$ linear potentiometers simulate analog sensor inputs:

- **POT1 (Voltage Fault Simulator):** Wiper output wired directly to GPIO33 (ADC1).
- **POT2 (Current Fault Simulator):** Wiper output wired directly to GPIO36 / VP (ADC1).
- **POT3 (Temperature Fault Simulator):** Wiper output wired directly to GPIO39 / VN (ADC1).
- **Electrical Safety Constraints:** The outer terminals of all three potentiometers are connected strictly between the 3.3 V supply rail and Common GND. **Never connect the high side of the simulation potentiometers to the 5V rail.** Rotating the potentiometer knobs varies the wiper voltage linearly from 0.0 V to 3.3 V , which internal scaling firmware maps to simulated $0\text{--}100\%$ parameter inputs.

Target UART Telemetry Streaming Protocol Layout

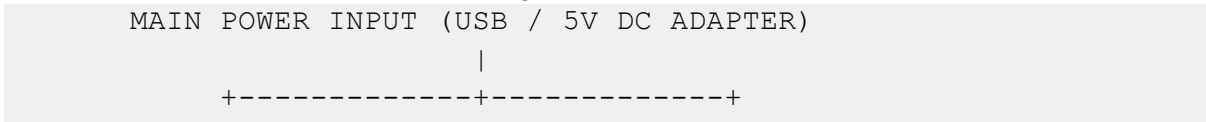
In the planned target architecture, the ESP32 packages normalized parameter percentages into a 5-byte telemetry frame transmitted over UART2 (115200 baud, 8 data bits, no parity, 1 stop bit) to the secondary safety decision engine:

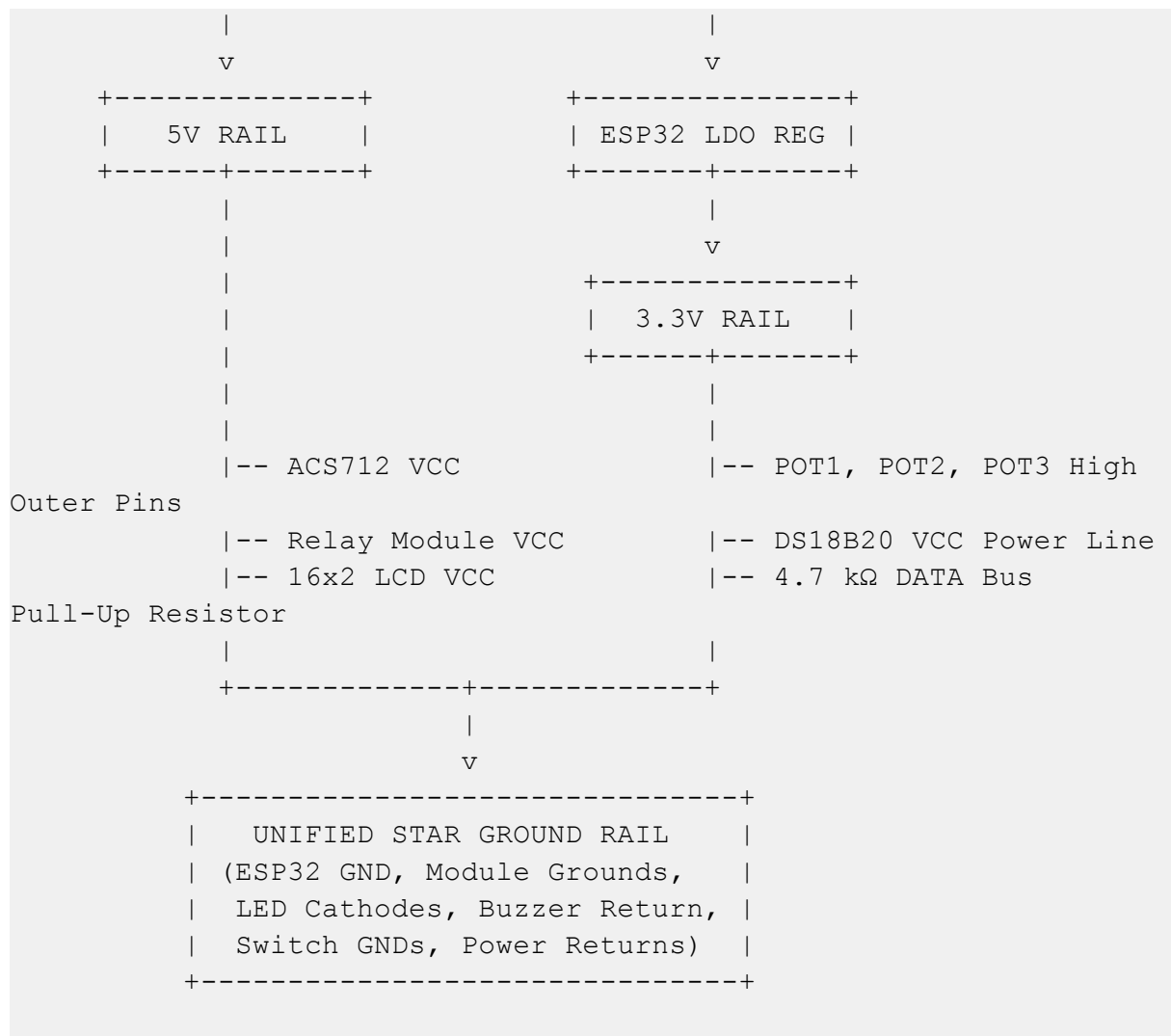
Frame Byte Index	Field Description	Range / Encoding	Function & Verification Rules
0	Synchronization Header	0xA5 (Fixed Hex Value)	Identifies frame start. The target parser discards bytes until 0xA5 is locked.
1	Voltage Parameter (V)	0x00 to 0x64 ($0\text{--}100\%$)	Normalized voltage percentage derived from 12-bit ADC conversion.
2	Current Parameter (I)	0x00 to 0x64 ($0\text{--}100\%$)	Normalized current percentage derived from 12-bit ADC conversion.
3	Temperature Parameter (T)	0x00 to 0x64 ($0\text{--}100\%$)	Normalized temperature percentage derived from ADC/1-Wire reading.
4	XOR Frame Checksum	0x00 to 0xFF	Calculated as: $\text{Byte 1} \oplus \text{Byte 2} \oplus \text{Byte 3}$. Frame dropped if invalid.

Robust sensor interfacing must be complemented by a clean, isolated power infrastructure to prevent noise injection and reference drift.

4. Power Infrastructure, Rail Isolation & Safety Protocols

In mixed-signal safety systems, rigorous power rail segregation and precise ground reference management are vital. Inductive transients generated during relay coil actuation, coupled with digital switching noise from microcontroller cores, can induce ground bounce and voltage fluctuations. These fluctuations introduce severe jitter into 12-bit ADC readings and can cause frame corruptions on single-ended UART links.





To maintain complete electrical integrity, the hardware infrastructure separates supply lines across three distinct functional domains: | Power Rail | Connected Subsystems & Components | Safety Directives & Structural Constraints || ----- | ----- | ----- || **3.3V Rail** | ESP32 LDO regulator output, POT1/POT2/POT3 outer pins, DS18B20 power pin, $4.7\text{ k}\Omega$ 1-Wire pull-up. | **Dedicated Analog Reference:** Supplies low-noise reference power. **Never connect potentiometers or sensor signal pull-ups to 5V.** || **5V Rail** | ACS712 Current Sensor V_{CC} , 1-Channel Relay Module V_{CC} , 16x2 I2C LCD V_{CC} . | **High-Current / Switching Rail:** Supplies high-current loads, display backlights, and inductive relay coils. Keep isolated from analog lines. || **Common GND** | ESP32 GND, ACS712 GND, Relay GND, LCD GND, Sensor GND, LED Cathodes, Buzzer negative, Switch terminals. | **Star Ground Architecture:** All ground returns terminate at a single physical point to eliminate ground loops, ADC offset drift, and UART corruption. |

Catastrophic ESD & Over-Voltage Failure Mode Analysis

Connecting 5.0 V powered inputs or placing potentiometer high-side pins on the 5 V rail exposes the ESP32 internal silicon to severe over-voltage conditions. The internal ESD clamp diodes on ESP32 GPIOs are rated to absorb transient events up to $V_{DD} + 0.3\text{ V}$ (3.60 V). Continuous exposure to voltages exceeding 3.6 V forces these clamp diodes into continuous conduction, causing high current

flow through the substrate. This results in thermal latch-up, permanent loss of ADC channel linearity, or permanent destruction of the GPIO matrix.

High-Voltage Isolation & Low-Voltage Test Mandate

- **CRITICAL SAFETY DIRECTIVE: NEVER CONNECT 230V AC MAINS VOLTAGE TO THE BREADBOARD, ESP32, SENSOR MODULES, OR RELAY CONTACTS.** This system is engineered exclusively for low-voltage DC test environments.
- **Isolated Low-Voltage Relay Path Architecture:** Power path switching must be isolated within a low-voltage DC loop:
- Low-Voltage Test Supply Positive (Source +) $\rightarrow$ Relay COM (Common Terminal).
- Relay NO (Normally Open Terminal) $\rightarrow$ Test Load Positive (Test Load +).
- Test Load Negative (Test Load -) $\rightarrow$ Low-Voltage Test Supply Return (Source -). Using the COM and NO terminals ensures that if V_{supply} drops, the control signal disconnects, or system logic halts, the internal coil de-energizes. Mechanical spring return pulls the contacts apart, physically isolating the load.

Decoupling and Transient Protection Guidelines

1. **ADC High-Frequency Filtering:** 100 nF ceramic surface-mount decoupling capacitors must be placed directly between ground and each analog input pin (GPIO32, GPIO35, GPIO33, GPIO36, GPIO39) as close to the ESP32 silicon as physically possible. These capacitors form passive RC low-pass filters that suppress ambient high-frequency noise.
2. **Bulk Supply Decoupling:** A $1000\text{ }\mu\text{F}$ electrolytic capacitor must be installed across the 5 V rail near the ESP32 board, and a second $1000\text{ }\mu\text{F}$ electrolytic capacitor must be placed directly across the supply terminals of the relay module. These capacitors absorb voltage drops caused by current spikes during relay coil energization.
3. **Inductive Transient Suppression:** Integrated relay modules include an onboard surface-mount flyback diode across the relay coil terminals. If a bare electromechanical relay is integrated, a 1N4007 power diode must be connected in reverse bias directly across the coil terminals (cathode to 5 V , anode to switching transistor collector) to clamp inductive voltage spikes generated during de-energization. With a stable and isolated power distribution network in place, the system executes real-time fault detection and state management via visual, audible, and relay outputs.

5. Visual/Audible Indicators & System State Logic

The ESP32 safety layer processes sensor readings into localized operator status indications while driving the power-path isolation relay based on real-time parameter validation.

```

+-----+
|  SENSOR EVALUATION  |
| (Highest % Input)   |
+-----+
|
+-----+

```

(80 - 100%)	(0 - 49%)	(50 - 79%)	
v		v	v
+-----+		+-----+	
+-----+			
SAFE STATE	WARNING STATE	FAULT STATE	
Green LED: ON	Green LED: OFF	Green LED: OFF	
Yellow LED: OFF	Yellow LED: ON	Yellow LED: OFF	
Red LED: OFF	Red LED: OFF	Red LED: ON	
Buzzer: OFF	Buzzer: OFF	Buzzer: ON	
Relay: ON (HIGH)	Relay: ON (HIGH)	Relay: OFF (LOW)	
+-----+	+-----+		
+-----+			

Actuator & Display Hardware Wiring

- **Status LEDs:** Visual indication relies on three distinct status LEDs. The long anode leg of each LED connects through an individual Ω series resistor to its designated control pin (GPIO2 for Green, GPIO4 for Yellow, GPIO15 for Red). Cathodes attach directly to Common GND.
- **Audible Alarm Subsystem:** The positive (+) terminal of the piezo buzzer connects directly to GPIO25, while the negative (-) terminal connects to Common GND. Driving GPIO25 HIGH (3.3 V) generates a continuous alarm tone; driving it LOW (0 V) silences the device.
- **Relay Module Control:** The control signal line (IN) connects directly to GPIO26. V_{CC} connects to 5 V , and GND connects to Common GND. Tested control logic: driving GPIO26 HIGH energizes the relay coil (closing COM to NO contacts to permit load charging); driving GPIO26 LOW de-energizes the coil (opening contacts to drop the load).
- **16x2 I2C LCD Interface:** The display backpack connects via 5 V V_{CC} , Common GND, SDA to GPIO21, and SCL to GPIO22. Verified hardware bus address is 0x27.

System State Machine & Threshold Mapping

Operational status is determined by taking the maximum normalized percentage across all three monitored parameter channels (Voltage, Current, Temperature): $\text{Max Parameter \%} = \max(\text{Voltage \%}, \text{Current \%}, \text{Temperature \%})$ | Operational State | Max Channel Range | Green LED (GPIO2) | Yellow LED (GPIO4) | Red LED (GPIO15) | Piezo Alarm (GPIO25) | Relay Control (GPIO26) | Charging Path Output || ----- | ----- | ----- | ----- | ----- | ----- || **SAFE** | $0\% \leq \text{Max} < 50\%$ | **ON** | OFF | OFF | OFF | OFF | **HIGH (ON)** | Energized (Closed) || **WARNING** | $50\% \leq \text{Max} < 80\%$ | OFF |

ON | OFF | OFF | **HIGH (ON)** | Energized (Closed) || **FAULT** | 80% | 100% | OFF | OFF | **ON** | **HIGH (ON)** | **LOW (OFF)** | **De-energized (Open)** |

Target FPGA State Machine Specification & Advanced Protections

In the planned target deployment, the hardware Verilog FSM executing on the PYNQ-Z2 FPGA implements four advanced functional safety features beyond basic software threshold checks:

1. **Watchdog Timeout Counter:** A 1000 ms (1.0 second) hardware watchdog timer runs continuously in Verilog. Every valid UART frame (correct 0xA5 sync header and matching XOR checksum) resets the watchdog counter. If frame transmission halts or becomes corrupted for longer than 1000 ms , the watchdog times out, forcing the safety FSM into an immediate FAULT state and opening the relay.
2. **Latched Fault Recovery:** Upon entering a FAULT state, the FSM latches the relay in the open position (LOW). Exiting the latched FAULT state requires two concurrent conditions: a manual pulse on the RESET push button *and* verification that all sampled parameters have dropped back into the safe operating zone.
3. **5% Hysteresis Band:** State transitions incorporate a 5-point (5%) hysteresis deadband below boundary entry thresholds. A state transition from WARNING back to SAFE occurs only when the maximum parameter drops below 45% (50% - 5%). A transition from FAULT back to WARNING requires parameters to drop below 75% (80% - 5%). This prevents output chatter and relay oscillation near decision boundaries.
4. **Power-On Safe State:** Upon power application or hardware reset, the relay defaults to a de-energized (LOW) open state. The relay energizes (HIGH) only after the first valid UART frame arrives and confirms that all parameters reside within the SAFE operating range.

Proposed Physical Thresholds for Target Build

When operating with physical transducers, normalized values map to real physical units:
Monitored Parameter | Physical Sensor | Measurement Range | Target Warning Limit | Target Fault Limit || ----- | ----- | ----- | ----- | ----- || **Temperature** | LM35 / DS18B20 | 0 to 100°C | 45.0°C | 60.0°C || **Charging Current** | ACS712-5A | 0.0 to 5.0 A | 3.0 A | 4.0 A || **Supply Voltage** | 0–25V Module | 0.0 to 25.0 V | 5.5 V | 6.0 V |

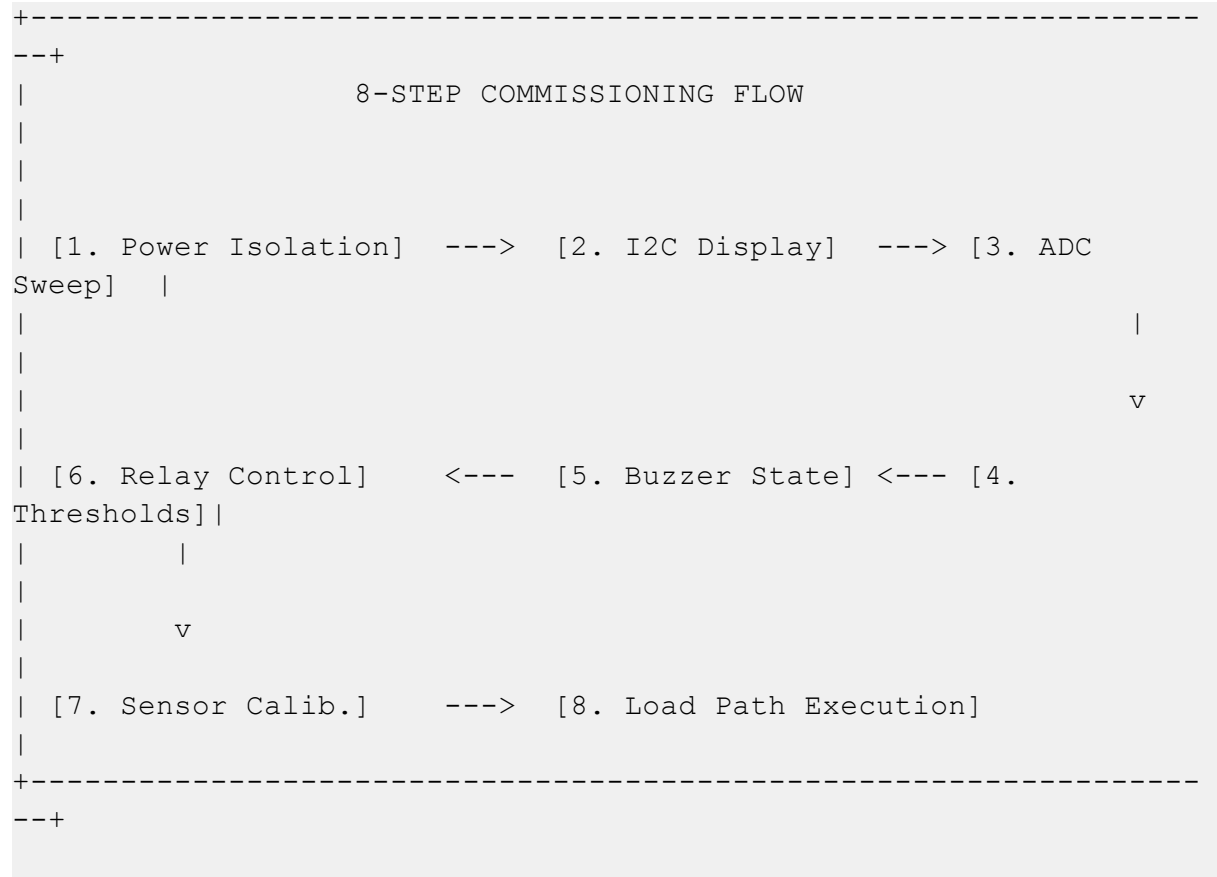
LCD Display Formatting Requirements

The 16x2 character LCD updates live system metrics on a 500 ms refresh cycle formatted as follows:

- **Line 1 Display Format:** V:xx% C:xx% (Renders live normalized Voltage and Current percentages).
- **Line 2 Display Format:** T:xx% SAFE (Renders live normalized Temperature percentage alongside the active system state string: SAFE, WARNING, or FAULT). Before deploying or operationalizing the hardware safety layer, a rigorous low-voltage verification protocol must be executed.

6. Low-Voltage Verification Protocols & Pre-Flight Checklist

To ensure circuit integrity, validate state machine response, and prevent component damage prior to deployment, the following 8-step bring-up sequence must be performed.



Step-by-Step Commissioning Procedure

- **Power Isolation Verification:** Apply power to the ESP32 strictly via USB connection. Ensure high-current test supplies are completely disconnected. Measure voltage across the 3.3V and 5V breadboard power rails using a calibrated digital multimeter to confirm supply stability.
- **I2C Display Initialization:** Power up the system and confirm that the 16x2 LCD backlight illuminates, initializes without line corruption, locks to address 0x27, and displays the default status text across Line 1 and Line 2.
- **Analog Simulation Input Sweep:** Individually rotate each simulation potentiometer (POT1, POT2, POT3) through its full mechanical travel. Confirm that the corresponding parameter percentages (0% to 100%) update dynamically on the LCD screen.
- **Threshold Zone Transition Validation:** Sweep the simulation potentiometers across defined input ranges to verify state machine transitions:
 - Maintain all inputs below 50% to confirm the Green LED illuminates, the Yellow and Red LEDs remain dark, and the LCD reads SAFE.
 - Sweep one input into the 50% to 79% range to verify illumination transitions exclusively to the Yellow LED (WARNING).
 - Sweep any single input to 80% or above to verify immediate illumination of the Red LED (FAULT).

- **Audible Alarm Functional Test:** Verify that the piezo buzzer connected to GPIO25 remains completely silent during SAFE and WARNING states, and generates a continuous audible alarm upon entering FAULT.
- **Relay Control Voltage Check:** Measure the control output logic on GPIO26 using a multimeter or oscilloscope. Confirm GPIO26 drives a HIGH logic signal (3.3 V) during SAFE and WARNING states, and immediately drops to LOW logic (0.0 V) upon entering a FAULT condition.
- **Standalone Sensor Calibration:** Prior to connecting physical transducers to active loads, validate the ACS712 current sensor and 0–25V voltage sensor independently against known low-voltage DC sources using a calibrated bench multimeter to establish accurate software scaling factors.
- **Load Path Shedding Execution:** Connect an isolated low-voltage DC source ($<12\text{ V}$) and dummy test load across the relay COM and NO terminals. Adjust a simulation potentiometer to force a FAULT state, and confirm physical load shedding via mechanical relay coil de-energization.

Pre-Flight Hardware Wiring Checklist

Checklist Verification Item, Expected Hardware State / Wiring Configuration, Verification Sign-Off

Star Grounding Unified, "ESP32 GND, ACS712 GND, Relay GND, LCD GND, sensor grounds, LED cathodes, buzzer negative, and switch grounds joined at a single physical point.", PASS

3.3V Power Rail Isolation, "POT1, POT2, POT3 outer pins, DS18B20 power, and $4.7\text{ k}\Omega$ pull-up tied exclusively to 3.3 V rail. 5.0 V connection strictly absent.", PASS

Analog Pin Allocations, "POT1 wiper to GPIO33, POT2 wiper to GPIO36 (VP), POT3 wiper to GPIO39 (VN). Inputs assigned exclusively to ADC1.", PASS

DS18B20 Interface Wiring, "Red wire to 3.3 V , Black to GND, DATA to GPIO19. $4.7\text{ k}\Omega$ resistor connected between GPIO19 and 3.3 V rail. DATA line unattached to 5 V .", PASS

ACS712 Attenuation Network, "Sensor V_{CC} to 5 V , GND to GND. Sensor OUT pin connected to $10\text{ k}\Omega$ series resistor; central node tied to GPIO35 and $20\text{ k}\Omega$ shunt to GND.", PASS

0–25V Sensor Header Config, "Screw terminals connected across load; Header pin S connected to GPIO32, Header - to GND, Header + left strictly unconnected.", PASS

I2C Bus Communication, "LCD V_{CC} to 5 V , GND to GND, SDA to GPIO21, SCL to GPIO22. Communication confirmed at I2C address 0x27.", PASS

LED Series Protection, "GPIO2 (Green), GPIO4 (Yellow), and GPIO15 (Red) connected through individual $220\text{ }\Omega$ current-limiting resistors prior to LED anodes.", PASS

Relay Drive Logic Setup, "Relay module V_{CC} to 5 V , GND to GND, IN to GPIO26. Confirmed active-HIGH logic configuration (HIGH = Relay ON / Closed).", PASS

Push Button Pull-Up Config, "Switches wired between GND and GPIO27 (START), GPIO14 (RESET), GPIO18 (FAULT), GPIO23 (MENU). Firmware set to INPUT_PULLUP.", PASS

Mains AC Complete Absence, 230V AC Mains voltage completely isolated and absent. Setup operates strictly using isolated low-voltage DC power sources., PASS